

Wednesday Warm Up: Unit 7: Forces V and Uniform Circular Motion, II

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1 Memory Bank

- $\tau = \vec{r} \times \vec{F}$... Definition of torque.
- $W = \vec{F} \cdot \Delta \vec{x}$... Definition of work (linear).
- $W = \vec{\tau} \cdot \Delta \vec{\theta}$... Definition of work (angular).
- $\tau = I\alpha$... Magnitude of work, angular acceleration, and moment of inertia, I .
- $I = mr^2$... Moment of inertia of a point mass m rotating with radius r .
- **For systems in static equilibrium, (1) the net force is zero, and (2) the net torque is zero.**
- $\vec{L} = \vec{r} \times \vec{p}$... Definition of *angular momentum*.
- $L = I\omega$, and $\vec{L} = I\vec{\omega}$, and $\Delta \vec{L} / \Delta t = I\vec{\alpha}$

2 Dot Product, and Cross Product

1. Find the *angular momentum*:

- $\vec{r} = -2\hat{i} + \hat{j}$ m, $\vec{p} = 4\hat{i} + \hat{j}$ kg m s⁻¹.
- $\vec{r} = 2\hat{i} + 1\hat{j}$ m, $\vec{p} = 1\hat{i} - 2\hat{j}$ kg m s⁻¹.

3 Torque and Angular Momentum

1. A motorcycle wheel has a mass of 12.0 kg. Assume it is a disk of radius 0.330 m. ($I = \frac{1}{2}MR^2$). The motorcycle is on its center stand, so that the wheel can spin freely. (a) If the drive chain exerts a force of 3000 N at a radius of 6.00 cm, what is the angular acceleration of the wheel? (b) What is the tangential acceleration of a point on the outer edge of the tire? (c) How long does it take to reach 80.0 rad/s, starting from rest? What is this in rpm?

2. Once the wheel has reached 80 rad/s, what is the angular momentum?

3. Suppose a loose part of the wheel fell off while the system is spinning at 80 rad/s. If the mass drops by 20%, what is the new angular velocity?

4. Remember the example of *las bolas*, the two balls spinning around a common radius, connected by a chord. (a) Show that the angular momentum is $L = 2mr^2\omega$, where m is the mass of one *bola*, r is the radius, and ω is the angular velocity.